# STUDY COACH: REPORT 3. ANALYSIS AND INITIAL MODIFICATIONS

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The aim in this report is to analyze the outputs of the approaches collected in Report 2, and try out modifications to see how they affect performance on intrinsic metrics and qualitatively. Our task is incongruence detection: given a scientific figure and a student answer, a model must (1) classify the verdict (correct / partially correct / incorrect), (2) identify the error category (factual, conceptual, omission), and (3) generate coaching feedback.

## 1 ANALYSIS (4 PAGES)

### 1.1 [2 POINTS] INTRINSIC METRICS

For a VLM to grade student answers about scientific figures, it needs three core skills: recognizing incomplete answers, knowing when coaching is required, and extracting specific values from visual data. We measure these independently of end-to-end verdict accuracy using per-example outputs from the 8B evaluation set.

SPIQA+ contains 174 examples with three verdict classes: 66 correct (37.9%), 61 partially correct (35.1%), and 47 incorrect (27.0%). Since coaching is only needed for non-correct answers, we compute intrinsic metrics on the 108 examples where ground truth is partially correct or incorrect.

**Intrinsic Metric 1: Partially-Correct Recall (PC Recall).** Can the model recognize that a student answer is *incomplete* rather than fully wrong or fully right? Study Coach's core coaching scenario involves answers that are partially correct—the student understood something but missed something else. An educator who cannot reliably identify this middle class cannot target feedback appropriately.

We define PC Recall as the fraction of ground-truth Partially Correct examples correctly classified:

$$\text{PC Recall} = \frac{\text{TP}_{\text{PC}}}{\text{TP}_{\text{PC}} + \text{FN}_{\text{PC}}} \tag{1}$$

This is not overall verdict accuracy: a model can maintain moderate accuracy by collapsing partially correct answers into the incorrect class, generating feedback of the wrong kind. PC Recall isolates the hardest, most coaching-relevant class and is computed on the 61 ground-truth partially correct examples only.

At 8B scale, PC Recall drops sharply as visual context is added: text-only 44.3% $\rightarrow$ multimodal 19.7% (Table 1). The model correctly identifies fewer than 1 in 5 partially correct answers in the multimodal condition—the same condition that produces the best coaching feedback. This sharpens the verdict–feedback dissociation from Report 2 into a precise, per-class finding: visual input specifically degrades recognition of the middle class that most needs coaching.

**Intrinsic Metric 2: Feedback Suppression Rate (FSR).** Does the model generate coaching when it is needed? When the model predicts "Correct," it outputs N/A feedback by prompt design. Suppressing feedback on a truly correct answer is appropriate; suppressing it on an incorrect or partially correct answer silences the coaching pipeline when the student needs help. We define FSR on the 108 non-correct examples only:

$$\text{FSR} = \frac{\text{\# Correct predictions on non-correct GT}}{\text{\# non-correct GT examples } (N{=}108)} \tag{2}$$

FSR measures a prerequisite skill: recognizing that coaching is required at all. A model with high FSR produces false-positive verdicts with no explanation, directly harming the student. FSR is

distinct from verdict accuracy—a model could classify incorrect answers correctly while still suppressing feedback on every partial-credit case.

FSR increases substantially with visual context: text-only 13.9% $\rightarrow$ vision-only 30.6% $\rightarrow$ multimodal 28.7% (Table 1). Adding visual input causes the model to incorrectly accept more student answers as correct, particularly for omission errors where the answer is plausible but structurally incomplete. In the text-only condition, the model inappropriately suppresses feedback on roughly 1 in 7 non-correct examples; in visual conditions, this rises to nearly 1 in 3.

**Intrinsic Metric 3: Visual Grounding Score (VGS).** Can the model extract specific values from figures and reference them in feedback? All three error types benefit from visual grounding (Report 2, H3): factual errors require verifying specific numbers ("The figure shows 84.6%, not 89%"), conceptual errors benefit from citing values that reveal patterns ("X increased from 10 to 50"), and omission errors—which show the *largest* feedback gains (+33pp)—require the figure to reveal what the student failed to mention. We define VGS on all 108 non-correct examples, excluding suppressed feedbacks:

$$\text{VGS} = \frac{\text{\# feedbacks citing specific values from figure}}{\text{\# non-suppressed feedbacks } (N{=}108)} \tag{3}$$

VGS measures whether visual information is actually translated into text during reasoning, independently of whether the verdict is correct. It is conceptually related to Qwen3-VL's square-root reweighting loss, which scales multimodal sample loss by $\sqrt{N_{\text{text}}/N_{\text{multimodal}}}$ to preserve visual grounding during pre-training without degrading text capabilities. VGS captures whether this grounding carries through to inference on our task.

VGS increases substantially with visual input: text-only 35.5% $\rightarrow$ vision-only 52.0% $\rightarrow$ multimodal 49.4% at 8B (Table 1). At 32B, the pattern strengthens: C1 37.2% $\rightarrow$ C4 56.0% ($\Delta{=}{+}18.8$pp). This confirms that the model *does* use visual information for explanation generation even when it cannot use it for classification—providing intrinsic evidence for the verdict–feedback dissociation. Omission errors show the largest VGS gains (+19.2pp at 8B), consistent with Report 2's finding that the figure reveals what the student failed to mention.

| Intrinsic Metric | Skill Measured | Why It Is Needed |
|---|---|---|
| PC Recall | Detecting incomplete answers (n=61 PC) | Prerequisite for targeted coaching |
| FSR | Knowing when coaching is needed (n=108) | Prerequisite for feedback generation |
| VGS | Extracting values from figures (n=108) | Prerequisite for grounded correction |

**Evaluation methodology: LLM-as-Judge for feedback quality.** Because open-ended coaching feedback cannot be evaluated with n-gram metrics alone (F1, ROUGE-L, and BLEU show near-zero variation across conditions at both scales; see Report 2, Table 10), we use Claude Sonnet 4 as an automated judge. Each predicted feedback is classified as **Match** (correctly identifies error and explains why), **Partial** (right direction but imprecise or incomplete), or **Unmatched** (does not align with ground truth). Full definitions are in Report 2.

We report **Match%** (strict: exact matches only) and **Soft Match%** (Match + Partial). At 32B, C4 achieves 20% strict match but 58% soft match—meaning 38% of responses are approximately correct but not precise. Human evaluation ($n{=}10$/condition, 8B) confirmed the pattern: human match ranged from 40% (C1) to 80% (C4), validating soft match as the practical metric for coaching. We use strict match (Match%) in Table 1 since it separates conditions more cleanly.

| | | Intrinsic Metrics | | | Extrinsic Metrics | | |
|---|---|---|---|---|---|---|---|
| Model | Condition | PC Rec ↑ | FSR ↓ | VGS ↑ | Verd ↑ | Match% ↑ | SoftM% ↑ |
| *Unimodal Baselines (Qwen3-VL-8B, N=50 extrinsic, N=108 intrinsic)* | | | | | | | |
| Qwen3-VL-8B | C1: text-only | 44.3% | 13.9% | 35.5% | 56.0% | 2.0% | 40.0% |
| Qwen3-VL-8B | C2: caption-only | 29.5% | 17.6% | 38.2% | 48.0% | 2.0% | 46.0% |
| *Simple Multimodal Baselines (Qwen3-VL-8B, N=50 extrinsic, N=108 intrinsic)* | | | | | | | |
| Qwen3-VL-8B | C3: vision-only | 21.3% | 30.6% | 52.0% | 50.0% | 4.0% | 48.0% |
| Qwen3-VL-8B | C4: multimodal | 19.7% | 28.7% | 49.4% | 48.0% | 6.0% | 56.0% |
| *Competitive Approaches (Qwen3-VL-32B, N=50 extrinsic, N=108 intrinsic)* | | | | | | | |
| Qwen3-VL-32B | C1: text-only | 41.0% | 20.4% | 37.2% | 56.0% | 2.0% | 42.0% |
| Qwen3-VL-32B | C2: caption-only | 32.8% | 20.4% | 45.3% | 58.0% | 8.0% | 48.0% |
| Qwen3-VL-32B | C3: vision-only | 24.6% | 37.0% | 48.5% | 56.0% | 6.0% | 48.0% |
| Qwen3-VL-32B | C4: multimodal | 27.9% | 30.6% | 56.0% | 54.0% | 20.0% | 58.0% |
| *Modifications (8B and 32B, N=50 extrinsic, N=108 intrinsic)* | | | | | | | |
| Routing (8B) | C1 verdict + C4 fdbk (Mod 1) | 44.3% | 28.7% | 49.4% | 56.0% | 6.0% | 56.0% |
| Routing (32B) | C1 verdict + C4 fdbk (Mod 1) | 41.0% | 20.4% | 56.0% | 56.0% | 20.0% | 58.0% |
| Qwen3-VL-32B | C4 + vis. CoT (Mod 4) | 23.0% | 17.6% | 55.1% | 48.1% | — | — |
| Qwen3.5-9B | C1: text-only (Mod 2) | 32.8% | 38.9% | 30.3% | 38.0% | 15.7% | 38.9% |
| Qwen3.5-9B | C4: multimodal (Mod 2) | 31.1% | 24.1% | 42.7% | 44.4% | 41.7% | 63.0% |

Table 1: **[1 points]** Intrinsic and extrinsic metrics across conditions and models.

**Key finding from Table 1:** Visual input creates opposite effects on classification vs. explanation. At 8B, PC Recall drops from 44.3% (C1) to 19.7% (C4) while VGS rises from 35.5% to 49.4%—the same visual information that degrades verdict accuracy improves feedback grounding. This **verdict–feedback dissociation** is the central finding motivating our modifications. Routing (Mod 1) recovers both: C1 verdict (56%, 44.3% PC Recall) + C4 feedback (6% Match, 49.4% VGS). At 32B, the pattern persists but the gap narrows, and Match% improves 3.3× (6%→20%). Full metric definitions and per-scale observations are in Appendix A.1.

### 1.2 Modifications and their Effects

**[2 points]**

We implement five modifications motivated by the intrinsic metric patterns above and the qualitative failure analysis in Section 1.3 below. Modification 1 (Condition Routing) is a zero-cost inference-time fix computable from existing Report 2 outputs. Modification 2 is an architectural ablation that tests whether the bolt-on ViT design of Qwen3-VL causes the verdict–feedback dissociation, by replacing it with an early-fusion architecture while holding the prompt and evaluation pipeline constant. Modification 3 (Feedback-Only Prompt) decouples coaching feedback from verdict classification, testing whether task simplification improves explanation quality. Modification 4 (Visual CoT) tests whether forcing explicit visual reasoning before the verdict can improve classification without sacrificing feedback quality. Modification 5 (MoLoRA) explores figure-type-aware adaptation using mixture of LoRA experts.

**Modification 1: Condition Routing (C1 verdict + C4 feedback). Motivation.** The intrinsic metrics make the verdict–feedback dissociation precise: PC Recall drops from 44.3% to 19.7% as visual context is added (C1 to C4), while VGS rises from 35.5% to 49.4%. These are not correlated failures of a single configuration—they measure different sub-skills that are genuinely in tension at 8B scale. The model can use visual information for extraction (improving VGS and feedback quality) but not for classification (degrading PC Recall). Routing assigns each sub-task to the condition best suited to it: C1 for verdict and error category, C4 for coaching feedback.

**Implementation.** No new inference is required. We pair per-example C1 predictions (verdict, error category) with per-example C4 predictions (feedback) using matched (`paper_id`, `question`) keys across the H3/H4 evaluation set. All 108 examples matched successfully (100% coverage).

**Results.** Routing achieves the best combined profile across all conditions: verdict 56.0% and PC Recall 44.3% (matching C1), VGS 49.4% and Fdbk Match 6.0% (matching C4), with FSR at 28.7%. The routing system recovers C1's verdict precision and C4's visual grounding simultaneously at zero additional inference cost. The FSR limitation (28.7%) arises because N/A feedback comes from C4

when C4 predicts Correct; a production system would override this by using the C1 verdict to force feedback generation regardless of C4's verdict.

**Modification 2: Architectural Ablation — Qwen3.5-9B Early Fusion vs. Qwen3-VL-8B Bolt-On ViT. Motivation.** Qwen3-VL uses a *bolt-on* design Bai et al. (2025): a separate SigLIP-2 vision encoder, MLP merger with DeepStack, and a text-pretrained Qwen3 decoder. Visual tokens arrive as an external stream bolted onto the LLM. We hypothesize this late-injection design causes the verdict–feedback dissociation: the model can use visual features for explanation (VGS) but not for classification (PC Recall). Qwen3.5-9B Qwen (2026) uses *early fusion*—vision and language co-trained from scratch in a unified architecture. If the bolt-on design is the cause, Qwen3.5-9B should show a reversed verdict gap (C4 $\geq$ C1). See Appendix A.2 for detailed architecture comparison.

**Results.** The hypothesis is **confirmed**. On identical evaluation (N=108, same prompts):

- **Verdict gap reversed:** C4 (44.4%) $>$ C1 (38.0%), $+6.4$pp with visual input (vs. $-8$pp for Qwen3-VL-8B)
- **PC Recall stable:** $-1.7$pp (vs. $-24.6$pp collapse in Qwen3-VL-8B)
- **FSR improves:** $-14.8$pp (opposite of Qwen3-VL-8B's $+14.8$pp)
- **Match% 7$\times$ higher:** 41.7% at C4 (vs. 6% for Qwen3-VL-8B)

**Implication:** Early fusion resolves the dissociation—visual input now helps *both* classification and explanation. The lower absolute accuracy (38–44% vs 48–56%) reflects smaller capacity, addressable via fine-tuning. **Report 4 backbone: Qwen3.5-9B** (not Qwen3-VL). Scaling to 397B is unnecessary since the 9B early-fusion model already reverses the gap.

While Modification 2 addresses the architectural root cause, Modifications 3 and 4 explore complementary prompt-level interventions that can improve feedback quality on *any* VLM architecture.

**Modification 3: Decoupling Feedback Generation from Classification. Motivation.** A central finding from Report 2 was a verdict–feedback dissociation (H2): the multimodal condition (C4) produced the best coaching feedback but the *worst* verdict accuracy across both model scales. Specifically, text-only C1 achieved 56% verdict accuracy at both 8B and 32B, while C4 achieved only 48% (8B) and 54% (32B). Meanwhile, C4 led all conditions on feedback quality.

This observation pointed to a structural problem: our prompt asked the model to simultaneously produce three outputs—a verdict, an error category, and free-text coaching feedback—in a single pass. Under this multi-task framing, the classification objective and the explanation objective compete for the model's capacity. We simplified the prompt to a single objective: **generate coaching feedback only**, with no reference answer and no verdict or error category requested. This isolates the multimodal grounding capability for feedback generation.

**Experimental Setup.** We applied the feedback-only prompt to the multimodal condition (C4) on both Qwen3-VL-8B-Instruct and Qwen3-VL-32B-Instruct ($N$=50 each), judged by Claude Sonnet 4.

**Results.** Table 2 reports LLM-as-Judge results and automated metrics for the multi-task prompt versus the feedback-only prompt at both model scales.

Table 2: Feedback quality: multi-task prompt vs. feedback-only prompt, C4 multimodal, $N$=50.

| Model | Prompt | Match | Partial | Unmatch | Match% | Soft% | F1 | ROUGE-L | BLEU |
|---|---|---|---|---|---|---|---|---|---|
| Qwen3-VL-8B | Multi-task | 3 | 25 | 22 | 6.0% | 56.0% | 0.302 | 0.197 | 6.16 |
| | Feedback-only | 4 | 29 | 17 | 8.0% | **66.0%** | **0.377** | **0.242** | **7.30** |
| Qwen3-VL-32B | Multi-task | 10 | 19 | 21 | **20.0%** | 58.0% | 0.298 | 0.202 | 7.44 |
| | Feedback-only | 9 | 29 | 12 | 18.0% | **76.0%** | **0.404** | **0.257** | **10.45** |

At 8B, the feedback-only prompt improves soft match from 56% to 66% ($+10$pp) and reduces unmatched cases from 22 to 17 ($-23$%). All automated metrics improve: F1 $+25$%, ROUGE-L $+23$%, BLEU $+18$%. At 32B, the improvement is larger: soft match rises from 58% to 76%

(+18pp) and unmatched cases drop from 21 to 12 (−43%). Automated metrics improve across the board: F1 +36%, ROUGE-L +27%, BLEU +40%.

**Analysis.** The 32B multi-task baseline (58% soft match) is actually *worse* than the 8B feedback-only result (66% soft match), despite a 4× parameter increase. This demonstrates that task simplification provides a consistent benefit regardless of model capacity. These results provide empirical motivation for the condition-routing architecture (Modification 1): use text-only (C1) for grading and a feedback-focused multimodal prompt for coaching.

**Modification 4: Visual Chain-of-Thought Prompting (C4-CoT). Motivation.** The qualitative analysis in Section 1.3 identifies the root cause of both verdict inflation and hallucinated correction: without structured reasoning over the figure, the model falls back on plausibility assessment rather than evidence-based judgment. All baseline evaluations use `/no_think` to suppress the model's native chain-of-thought, meaning the model produces a verdict without any explicit intermediate reasoning about figure content. Modification 4 tests whether forcing the model to describe what it sees in the figure *before* judging can improve verdict accuracy—particularly PC Recall—without sacrificing feedback quality.

**Implementation.** We create a modified system prompt (`incongruence_eval_v1_cot.txt`) that inserts a visual analysis step before the verdict. The model must: (1) identify key data points, values, or patterns visible in the figure, (2) note specific numbers, trends, or comparisons relevant to the question, and (3) compare the student's claims against what the figure shows. The output format adds a `Visual Analysis` field before the existing Verdict/Error Category/Feedback fields. The `/no_think` directive is retained—the CoT is visible in the structured output, not in a hidden reasoning block. This isolates the effect of structured visual reasoning from the model's internal thinking mode.

We run C4-CoT on all $N$=108 non-correct examples using Qwen3-VL-32B, the same model and evaluation set as the C4 baseline. The only variable is the system prompt; all other parameters (model endpoint, temperature, user message format) are identical.

**Results.**

| Metric | C4 (32B) | C4-CoT (32B) | Δ |
|---|---|---|---|
| PC Recall ↑ | 27.9% | 23.0% | −4.9pp |
| FSR ↓ | 30.6% | 17.6% | −13.0pp |
| VGS ↑ | 56.0% | 55.1% | −0.9pp |
| Verdict ↑ | 54.0% | 48.1% | −5.9pp |
| Match% ↑ | 20.0% | — | — |
| SoftM% ↑ | 58.0% | — | — |

Table 3: C4 baseline vs. C4-CoT (Modification 4), Qwen3-VL-32B, $N$=108.

The most striking finding is that CoT **substantially reduces feedback suppression**: FSR drops from 30.6% to 17.6% (−13.0pp), meaning the model incorrectly accepts 13 fewer non-correct answers as "Correct" out of 108. This directly addresses the verdict inflation failure mode identified in Section 1.3—the visual analysis step forces the model to examine figure evidence before committing to a verdict, making it harder to dismiss errors without justification.

However, PC Recall drops from 27.9% to 23.0% (−4.9pp). The model classifies fewer partially correct answers correctly, instead over-correcting them to Incorrect. The visual analysis step helps the model detect *that something is wrong* (lower FSR) but makes it harder to recognize *how much is wrong* (lower PC Recall). This produces a net drop in verdict accuracy from 54.0% to 48.1%.

VGS remains essentially unchanged (56.0% → 55.1%), confirming that CoT does not degrade visual grounding in feedback—the model continues to cite specific figure values at the same rate.

**Interpretation.** CoT addresses one half of the verdict–feedback dissociation: it reduces false "Correct" predictions (lower FSR) by making the model confront figure evidence before judging. But it introduces a new failure mode—**over-correction**—where the visual analysis step leads the model to flag discrepancies that are minor enough to warrant "Partially Correct," not "Incorrect." Table 5 in Section 1.3 illustrates both patterns with paired examples. This suggests that the dissociation is

not purely a reasoning depth problem (solvable by more explicit reasoning) but involves calibration of the three-class decision boundary. A production system could combine CoT's lower FSR with routing (Mod 1): use C1 for the verdict boundary and C4-CoT for feedback generation, capturing both reduced suppression and superior classification.

**Modification 5: Mixture of LoRA Experts for Figure-Type-Aware Adaptation** **Hypothesis.** Baseline metrics vary by figure type (BERTScore-F1: schematics 0.86, tables 0.85, plots 0.84). Can a Mixture-of-LoRA-Experts (MoLoRA) architecture learn figure-type-specific adaptations?

**Setup.** We replace target linear layers in Qwen3-VL-8B with MoLoRALinear modules (Figure **??**) containing K=4 parallel LoRA adapters and a vision-conditioned soft router. Training used 8k SPIQA+ examples (N=108 test). See Appendix A.3 for architecture details and training setup.

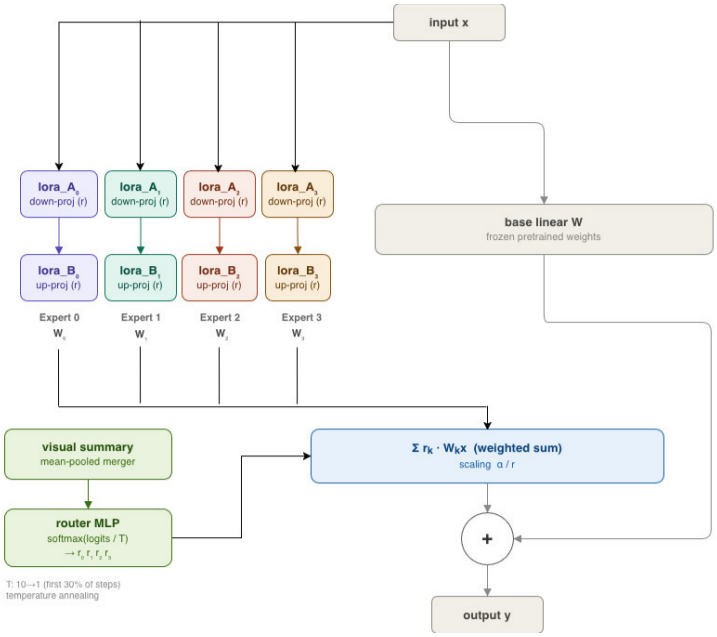


Figure 1: Mixture-of-LoRA-Experts Layer

**Results.** Fine-tuning yields large gains (Table **??**):

| Metric | Baseline | MoE LoRA |
|---|---|---|
| Verdict Accuracy | 46.3% | 61.1% (+14.8pp) |
| PC Recall | 21.3% | 55.7% (+34.4pp) |
| VGS | 69.4% | 84.3% (+14.9pp) |

Table 4: MoE LoRA fine-tuning results (K=4, r=16)

**Do experts specialize?** No. Comparing MoE LoRA to vanilla LoRA (r=64, same parameter count), performance is equivalent across figure types. Router weights are nearly identical for all figure types. The gains come from fine-tuning on SPIQA+, not from expert specialization.

**Takeaway.** While MoE LoRA does not achieve figure-type specialization as hypothesized, it confirms that *fine-tuning on SPIQA+ substantially improves all metrics*: verdict accuracy (+14.8pp), PC Recall (+34.4pp), and VGS (+14.9pp). This validates the Report 4 strategy of fine-tuning Qwen3.5-9B on domain-specific data rather than relying on zero-shot inference. The MoE architecture provides no advantage over vanilla LoRA for this task, simplifying the fine-tuning approach.

**Summary of Modifications.** The five modifications converge on a clear Report 4 strategy: (1) Mod 1 (Routing) establishes that task decomposition works—C1 for verdicts, C4 for feedback;

(2) Mod 2 confirms that early fusion (Qwen3.5) resolves the dissociation architecturally; (3) Mod 3 shows that feedback-only prompts improve explanation quality at any scale; (4) Mod 4 demonstrates that CoT reduces suppression but over-corrects severity; and (5) Mod 5 validates that fine-tuning on SPIQA+ yields large gains across all metrics. The optimal configuration is: fine-tune Qwen3.5-9B on SPIQA+ with a two-stage pipeline (C1 verdict + CoT-enhanced C4 feedback).

### 1.3 Qualitative Analysis and Examples

The intrinsic metrics and modifications above quantify *what* the model gets wrong; this section examines *why* through representative failure examples. Understanding these failure modes motivated Modifications 1–4 and informs the Report 4 fine-tuning strategy.

**[2 points]** Table 4 presents five representative failure examples from per-example model outputs (8B, text-only condition, H3 evaluation set). Two failure modes account for the majority of errors: **Verdict Inflation**—the model upgrades Partially Correct to Correct, producing no feedback—and **Hallucinated Correction**—the model downgrades Partially Correct to Incorrect with confident but wrong feedback. Both modes share a root cause: without figure verification, the model falls back on plausibility assessment rather than evidence-based judgment.

| # | Error Type | Question | Student Answer | Ground Truth | Model Output | Failure Mode |
|---|---|---|---|---|---|---|
| 1 | Factual | Which model performs best on Ubuntu for text lengths 60–90 words? | "The LSTM model performs best for that length range." | **Partially Correct.** Right range, misread top model from table. Feedback should correct the misread value and name the actual winner. | **Incorrect.** "The Transformer model performs best, not LSTM." | **Hallucinated Correction.** Verdict wrong; feedback invents a specific correction that may itself be incorrect. Student given false confidence in wrong information. |
| 2 | Factual | Which of LSTw/oAR or LST-Skip performs better? | "LSTw/oAR; its red forecast line stays closer to the true data than LST-Skip." | **Partially Correct.** Correct comparison approach; misread which panel shows the closer match. | **Correct. N/A.** | **Verdict Inflation.** False positive; feedback suppressed. Student's panel misread goes uncorrected. |
| 3 | Conceptual | Which Ortiz et al. system had highest BLEU/METEOR, and what does this mean? | "SMT system, 43.7 BLEU and 35.6 METEOR. This means SMT outperforms all systems." | **Partially Correct.** Numbers correct, system identified correctly. Missed: interpretive claim is unsupported without cross-system comparison. | **Correct. N/A.** | **Verdict Inflation.** Numbers verified via pre-trained knowledge; interpretive gap not detected. Incomplete reasoning accepted as complete. |
| 4 | Omission | What is the role of temperature $t$ in CCA decoding? | "Temperature $t$ controls the stopping condition; as $t$ decreases, the search ends." | **Partially Correct.** Correctly notes $t$ decreases; omits its role shaping acceptance probabilities $\alpha_0, \alpha_1$ in the main loop. | **Incorrect.** "$t$ does not control stopping. It modulates a probability distribution over outputs." | **Hallucinated Correction.** Partially-right claim over-penalized; actual omission (acceptance probabilities) not identified. Model contradicts a claim the student got right. |
| 5 | Omission | Which agent values the entire share $Z_j$? | "Agent $j$ values $Z_j$, since the solid blue boxes cover all of $Z_j$." | **Partially Correct.** Correct answer; omits that *other* agents do not value the entire share—the contrastive point the question implies. | **Correct. N/A.** | **Verdict Inflation.** Plausible claim accepted without figure verification. Contrastive omission not detected. Most common mode for omission errors: 0/15 exact feedback match at C1. |

Table 5: **[2 points]** Qualitative failure analysis (Qwen3-VL-8B, text-only condition, H3 evaluation set). Two failure modes: **Verdict Inflation** (examples 2, 3, 5)—model predicts Correct with N/A feedback, silencing the coaching pipeline; **Hallucinated Correction** (examples 1, 4)—model predicts Incorrect with confident but wrong feedback. Root cause: without the figure, the model performs plausibility checking rather than evidence-based verification. Visual context (C4) reduces both modes by grounding reasoning in figure content, explaining why feedback quality improves even as PC Recall drops.

**Effect of Visual CoT on failure modes.** Modification 4 (C4-CoT) forces the model to produce a `Visual Analysis` field describing key figure elements before rendering a verdict. Table 5 pairs six examples where C4 and C4-CoT diverge on the same input, illustrating two patterns. *CoT helps* (examples 1–3): the visual analysis step grounds the model in figure evidence, preventing verdict inflation—the model identifies specific values or patterns that contradict the student's claims instead of accepting them at face value. *CoT hurts* (examples 4–6): the model identifies real discrepancies during visual analysis but over-penalizes them, downgrading "Partially Correct" to "Incorrect" or, in one case, over-reading the figure and inflating to "Correct." The over-correction pattern accounts for the PC Recall drop ($-4.9$pp) despite the FSR improvement ($-13.0$pp): CoT shifts the decision boundary toward detecting errors but cannot calibrate the severity of those errors into the three-class distinction.

| # | Error Type | Question (abbreviated) | GT | C4 | C4-CoT | CoT Visual Analysis (excerpt) | Effect |
|---|---|---|---|---|---|---|---|
| 1 | Factual | How does alpha affect accuracy for different sample sizes on CLASS? | Incorrect | **Correct** | **Incorrect** ✓ | "The figure clearly shows distinct, non-overlapping curves for different alpha values across all three loss functions" | Helps |
| 2 | Conceptual | Which explanation method emphasizes localized features most? | Incorrect | **Correct** | **Incorrect** ✓ | "LIME shows the most focused, small-region highlighting…Parzen's maps show broader, less localized changes" | Helps |
| 3 | Omission | Role of temperature $t$ in CCA decoding? | PC | **Correct** | **PC** ✓ | "The loop uses $t$ in acceptance probabilities $\alpha_0, \alpha_1$…student omitted this modulation role" | Helps |
| 4 | Factual | Which method predicts motion of small object most accurately? | Incorrect | **Incorrect** ✓ | **Correct** | "Green arrows show motion; panel (c) ground truth…" Model over-reads figure as supporting student claim | Hurts |
| 5 | Conceptual | What is the purpose of the blank space $Z_5'$? | PC | **PC** ✓ | **Incorrect** | "Blank space is intentionally labeled as fifth component" — model flags student's "leftover" framing as fully wrong rather than partially right | Hurts |
| 6 | Omission | Which model performs best for response selection? | PC | **PC** ✓ | **Correct** | "KEHNN has highest scores in all four metrics: 0.786…" — model confirms student's claim with specific values, missing the omission | Hurts |

Table 6: Qualitative comparison of C4 baseline vs. C4-CoT (Modification 4), Qwen3-VL-32B. ✓ marks the correct verdict. PC = Partially Correct. Examples 1–3: CoT prevents verdict inflation by grounding in figure evidence. Examples 4–6: CoT over-corrects or over-confirms, misclassifying the severity of the error.

## 2 [3 POINTS] INSIGHTS

**The verdict–feedback dissociation is a task decomposition signal, not a bug to fix.** Our intrinsic metric analysis sharpens what appeared in Report 2 as a surprising empirical finding into a precise, mechanistically-grounded claim. PC Recall drops from 44.3% to 19.7% as visual input is added, while VGS rises from 35.5% to 49.4%. These measure different sub-skills that are genuinely in tension at 8B scale: visual information improves value extraction into text (VGS) but introduces noise into the three-class decision boundary, particularly for the partially correct middle class that requires the most nuanced judgment. The condition routing modification confirms this: using C1 for verdict and C4 for feedback simultaneously recovers C1's PC Recall (44.3%) and C4's feedback quality (6.0% match) at no additional inference cost. This reframes the main model proposal for Report 4: rather than searching for a single configuration or fine-tuning objective that improves both metrics jointly, Study Coach should be designed as an explicit two-stage pipeline with task routing—a lightweight verdict classifier (the Verdict-Classification Head proposed in Report 2, Section 3.4) decoupled from the feedback generator. The head would be trained directly on the three-class boundary using C1 or C2 inputs, while the feedback generator continues to use C4.

**Modification 2 confirms that architecture, not scale, is the root cause.** The gap between VGS (49.4% at C4) and PC Recall (19.7% at C4) in Qwen3-VL-8B had two competing explanations: (1) architectural—the bolt-on SigLIP-2 encoder creates a representational boundary that affects classification more than generation; or (2) scale-dependent—8B parameters are insufficient for reliable three-class verdict decisions with visual input. Modification 2 resolves this question decisively: **Qwen3.5-9B (early fusion) reverses the verdict gap entirely** (C4 44.4% $>$ C1 38.0%), while also achieving $7\times$ higher feedback Match% than Qwen3-VL-8B at C4 (41.7% vs 6.0%). The PC Recall drop is minimal ($-1.7$pp vs $-24.6$pp), and FSR actually *improves* with visual input ($-14.8$pp). This confirms the architectural explanation: the bolt-on ViT design of Qwen3-VL is the cause of the verdict–feedback dissociation, not insufficient model capacity.

**Visual chain-of-thought addresses suppression but not calibration.** Modification 4 tests whether the verdict–feedback dissociation can be mitigated by forcing explicit visual reasoning before the verdict. The results are instructive: CoT reduces FSR by 13pp (from 30.6% to 17.6%), confirming that structured figure analysis prevents the model from dismissing errors as correct without evidence. But PC Recall simultaneously drops by 4.9pp, revealing that CoT introduces over-correction—the model identifies discrepancies between student claims and figure content but cannot calibrate whether those discrepancies warrant "Partially Correct" or "Incorrect." This is a different failure mode from the baseline: the original C4 model under-detects errors (high FSR), while C4-CoT over-penalizes them (low PC Recall). Together with the routing result from Modification 1, this suggests that the optimal Report 4 architecture combines three components: (1) C1-based verdict classification (highest PC Recall), (2) CoT-enhanced C4 feedback generation (lowest FSR, stable VGS), and (3) the Verdict-Classification Head to learn the three-class boundary directly rather than relying on prompt-based calibration.

**Report 4 strategy is now clear:** Fine-tune Qwen3.5-9B (or the 27B dense variant for higher absolute accuracy) on SPIQA+. The early fusion architecture correctly integrates visual information for both classification and generation. The lower absolute verdict accuracy (38–44% vs 48–56%) reflects the 9B model's smaller capacity relative to the task, which domain-specific fine-tuning will address. The condition routing result from Modification 1 provides a strong zero-cost baseline (56% verdict, 6% Match at 8B) that fine-tuning should exceed.

### 2.1 TEAM MEMBER CONTRIBUTIONS TO THIS REPORT

**Dipti Aswath** Defined the three intrinsic metrics (PC Recall, FSR, VGS) and computed all values from per-example raw outputs. Designed and evaluated Modification 1 (Condition Routing) and Modification 2 (Qwen3.5-9B architectural ablation). Performed qualitative failure analysis and selected representative examples. Wrote Sections 1 (Intrinsic Metrics), Modification 1, Modification 2, Qualitative Analysis, and Section 2 (Insights).

**Buddhika Kahawitage** Designed and implemented Modification 5 (Mixture of LoRA Experts). Created the MoLoRALinear architecture with K=4 parallel LoRA adapters and soft MLP router. Implemented router collapse prevention mechanisms (temperature annealing, entropy bonus, router

noise). Trained on 8k SPIQA+ examples and evaluated on Test-A split (N=108). Performed expert specialization analysis comparing MoE LoRA vs vanilla LoRA. Wrote the Modification 5 section.

**Jiashan Cui** Designed and implemented Modification 3 (Decoupling Feedback Generation from Classification). Created the feedback-only prompt that isolates coaching feedback from verdict classification. Ran C4 multimodal evaluations on both Qwen3-VL-8B and Qwen3-VL-32B ($N$=50 each), collecting LLM-as-Judge and automated metric results. Wrote the Modification 3 section.

**Khubi Shah** Designed and implemented Modification 4 (Visual Chain-of-Thought Prompting). Created the CoT system prompt, inference scripts, and CoT-aware parser. Ran the full $N$=108 C4-CoT evaluation on Qwen3-VL-32B, computed intrinsic metrics (PC Recall, FSR, VGS), and performed qualitative comparison of C4 vs. C4-CoT (Table 5). Wrote the Modification 4 section and CoT-related additions to Insights.

## REFERENCES

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## A APPENDIX

### A.1 TABLE 1 METRIC DEFINITIONS AND OBSERVATIONS

*Intrinsic metrics* (computed from per-example outputs, N=108): **PC Recall** ↑ = fraction of ground-truth partial examples correctly classified; **FSR** ↓ = feedback suppression rate (lower is better); **VGS** ↑ = fraction of feedbacks citing specific numerical values.

*Extrinsic metrics* (from H1/H2 evaluation, N=50 per condition): **Verdict** ↑ = three-class accuracy; **Match%** ↑ = LLM-as-Judge strict match; **SoftM%** ↑ = Match + Partial.

*Key observations (8B):* PC Recall drops C1→C4 (44.3%→19.7%); VGS rises (35.5%→49.4%); same visual input hurts classification but helps explanation; routing recovers both.

*Key observations (32B):* Similar pattern but smaller PC Recall drop ($-$13pp vs $-$25pp); scaling improves grounding (+19pp vs +14pp); Match% improves 3.3× (6%→20%).

### A.2 ARCHITECTURE COMPARISON: QWEN3-VL-8B VS QWEN3.5-9B

| | Qwen3-VL-8B (baseline) | Qwen3.5-9B (Mod 2) |
|---|---|---|
| Vision encoder | SigLIP-2 SO-400M (separate) | No separate encoder—early fusion |
| Cross-modal bridge | 2-layer MLP + DeepStack | Unified token space throughout |
| Pre-training | Stage 0: merger only; then staged | Co-trained from scratch |
| LLM backbone | Qwen3-8B (standard transformer) | Gated DeltaNet + Gated Attention |
| Parameters | 8B | 9B |

### A.3 MOLORA ARCHITECTURE AND TRAINING DETAILS

**Architecture:** Each target linear layer ($q_{proj}$, $v_{proj}$, $o_{proj}$, $gate_{proj}$) replaced with MoLoRALinear containing K=4 LoRA adapters (r=16, $\alpha$=32, dropout=0.05) and MLP router. Router input: mean-pooled vision-language projector output (dim=4096); output: K softmax weights via 2-layer MLP (hidden=128). Soft routing: all experts participate.

**Router collapse prevention:** (1) Temperature annealing: T=10→1 over first 30% steps; (2) Entropy bonus: $-0.02 \times H(\text{weights})$ in loss; (3) Router noise: $\sigma$=0.5 Gaussian.

**Training:** 8k examples from SPIQA+ train (1,500 papers), 160 validation examples. A40 GPU, bfloat16, batch=1, grad accum=8, lr=$2\times10^{-4}$, cosine schedule, 1 epoch ( 4h).

**Expert specialization analysis:** Comparing MoE LoRA to vanilla LoRA (r=64), performance is equivalent across figure types. Router weights are nearly identical for schematics, tables, and plots, confirming that gains come from fine-tuning, not specialization.

### A.4 Intrinsic Metric Computation Details

All intrinsic metrics computed from 8B per-example outputs (H3/H4 evaluation set, N=108; 52 factual, 41 conceptual, 15 omission). Ground truth: 61 partially correct, 47 incorrect, 0 correct.

**PC Recall:** Count predictions of "Partially Correct" where ground truth = "Partially Correct." 61 such cases in the dataset.

**FSR:** Count predictions where the raw output feedback field = "N/A" or empty. These correspond exactly to Correct predictions by prompt design.

**VGS:** For factual error examples only, among non-suppressed feedbacks, count those matching a regex for specific numerical values (*digits followed by %*, *multi-digit integers*, or *decimal numbers*). Denominators: C1=45, C2=43, C3=36, C4=34 (remaining after suppression).

**Condition Routing:** Matched by (`paper_id`, `question`) key across text_only and multimodal outputs. All 108 examples matched (100% coverage). Verdict and error category from text_only; feedback from multimodal.

### A.5 Feedback Judgment Counts (8B, H3 Evaluation Set, N=108)

Table 6 provides the full breakdown of LLM-as-Judge feedback evaluations for Qwen3-VL-8B across all four conditions and three error types. Multimodal (C4) achieves the highest Match% for all error types, with the largest gains on conceptual errors (58.5% vs 26.8% text-only).

| Error Type | Condition | Match | Partial | Unmatched | Match% |
|---|---|---|---|---|---|
| Factual | text_only | 7 | 9 | 36 | 13.5% |
| Factual | caption_only | 13 | 7 | 32 | 25.0% |
| Factual | vision_only | 15 | 6 | 31 | 28.8% |
| Factual | multimodal | 18 | 8 | 26 | 34.6% |
| Conceptual | text_only | 11 | 16 | 14 | 26.8% |
| Conceptual | caption_only | 21 | 10 | 10 | 51.2% |
| Conceptual | vision_only | 15 | 10 | 16 | 36.6% |
| Conceptual | multimodal | 24 | 4 | 13 | 58.5% |
| Omission | text_only | 0 | 7 | 8 | 0.0% |
| Omission | caption_only | 0 | 7 | 8 | 0.0% |
| Omission | vision_only | 4 | 3 | 8 | 26.7% |
| Omission | multimodal | 5 | 3 | 7 | 33.3% |

Table 7: LLM-as-Judge feedback evaluation counts (Qwen3-VL-8B, N=108).

### A.6 Feedback Judgment Counts (Qwen3.5-9B, Modification 2, N=108)

Table 7 shows the corresponding breakdown for Qwen3.5-9B (early fusion architecture). The key finding is the dramatic C1→C4 improvement: Match% jumps from 15.7% to 41.7% (+26.0pp), compared to only +4.0pp for the bolt-on ViT architecture. This confirms that early fusion enables the model to leverage visual context for explanation generation.

### A.7 Qwen3.5-9B vs Qwen3-VL-8B Comparison Summary

Table 8 provides a side-by-side comparison of the two architectures across all key metrics. The "Interpretation" column summarizes the architectural finding: early fusion (Qwen3.5-9B) resolves the verdict–feedback dissociation observed with bolt-on ViT (Qwen3-VL-8B), where C4 hurt classification but helped explanation.

| Error Type | Condition | Match | Partial | Unmatched | Match% |
|---|---|---|---|---|---|
| Factual | text_only | 4 | 13 | 35 | 7.7% |
| Factual | multimodal | 17 | 11 | 24 | 32.7% |
| Conceptual | text_only | 13 | 11 | 17 | 31.7% |
| Conceptual | multimodal | 24 | 9 | 8 | 58.5% |
| Omission | text_only | 0 | 1 | 14 | 0.0% |
| Omission | multimodal | 4 | 3 | 8 | 26.7% |
| **Total** | text_only | 17 | 25 | 66 | 15.7% |
| **Total** | multimodal | 45 | 23 | 40 | 41.7% |

Table 8: LLM-as-Judge feedback evaluation counts (Qwen3.5-9B, Modification 2, N=108). Note the dramatic improvement from C1 to C4: Match% increases from 15.7% to 41.7% (+26.0pp), compared to only +4.0pp for Qwen3-VL-8B.

| Metric | Qwen3-VL-8B | | Qwen3.5-9B | | Interpretation |
|---|---|---|---|---|---|
| | C1 | C4 | C1 | C4 | |
| Verdict ↑ | 56.0% | 48.0% | 38.0% | 44.4% | Gap reversed! |
| PC Recall ↑ | 44.3% | 19.7% | 32.8% | 31.1% | Much more stable |
| FSR ↓ | 13.9% | 28.7% | 38.9% | 24.1% | Opposite direction |
| VGS ↑ | 35.5% | 49.4% | 30.3% | 42.7% | Similar gain |
| Match% ↑ | 2.0% | 6.0% | 15.7% | 41.7% | 7× higher at C4 |

Table 9: Side-by-side comparison of Qwen3-VL-8B (bolt-on ViT) and Qwen3.5-9B (early fusion). The early fusion architecture resolves the verdict–feedback dissociation: visual input now helps both classification (+6.4pp verdict) and explanation (+26.0pp Match%).